

A Player-Aware Machine Learning Approach for FIFA World Cup Match Outcome Prediction

Mini Project Report

11 - Hussain Cochinwala

15 - Aditya Dhuri

23 - Abhishek Joshi

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1 Abstract

Most football match prediction systems rely only on team-level statistics such as historical results and rankings, ignoring the fact that national squads are almost entirely rebuilt between World Cup cycles. Team reputation is therefore a weak and slow-moving signal for a tournament whose rosters change every four years. This project builds a **player-aware** match outcome predictor that fuses individual player attributes — technical, physical, mental, and goalkeeping ratings — aggregated into year-specific team profiles, with team-level match history computed *as of each match date* (Elo, weighted win ratio, head-to-head record, recent form). The resulting features are standardized and reduced via Principal Component Analysis (PCA), then fed into a five-model ensemble (Logistic Regression, Random Forest, XGBoost, AdaBoost, KNN) combined through hard majority voting, and benchmarked against a Weighted Win Ratio (WWR) baseline derived from historical results alone.

The dataset comprises **1,749 team-year squad profiles** (189 nations, FIFA editions 2015–2025) joined onto **1,504 usable World Cup matches** (finals and qualifiers, draws excluded), described by **64 engineered features** reduced to **22–23 principal components**. To avoid reporting a figure that model selection had already been run against, the **2025 season was withheld entirely** from development and scored exactly once. On that holdout the ensemble reaches **82.2% accuracy against the baseline’s 75.4%** — a **+6.8 percentage-point margin that is statistically significant** (McNemar’s test, $p = 0.0134$), with ROC-AUC 0.907. On development data (2015–2024, five random seeds) the ensemble scores **79.0% ± 3.3%** against **76.0% ± 3.1%**, ahead on all five seeds. For reference, the paper this work reproduces reports 59.38% against a 56.25% baseline. The full pipeline is exposed through a live web application (FastAPI backend + React frontend) giving per-match predictions with model agreement, alongside an interactive evaluation dashboard.

2 Introduction

Predicting the outcome of a football match is a well-studied but persistently hard problem: results are influenced by dozens of interacting factors, from squad quality to tournament pressure to raw luck. Two prediction paradigms dominate the literature:

1. **Team-level models** — use historical results, rankings (e.g. FIFA or Elo ratings), and head-to-head records as features. These are simple and reasonably strong for club football, where squads persist season to season.

2. **Player-level / player-aware models** — incorporate individual player attributes, aggregated to the team level, on top of team-level history.

For the FIFA World Cup specifically, team-level models face a structural weakness: national squads are drawn from players across dozens of different club leagues and are effectively rebuilt for every tournament cycle. A team’s historical win ratio reflects a *different* group of players than the one that will take the field next tournament. This motivates a player-aware approach, where the technical and physical attributes of the current squad are a primary signal and historical team form is a complementary one.

This project follows the methodology of Al-Bustami & Ghazal (2025), who propose exactly this combination — player attributes aggregated into year-specific team profiles, dimensionality-reduced via PCA, and classified with a five-model majority-voting ensemble — benchmarked against a Weighted Win Ratio baseline. We reproduce that pipeline end-to-end on real, public data, evaluate it under the same metrics, and extend it in three directions the original does not cover:

- **Information parity between the two models.** In the paper’s framing the ensemble sees squad attributes and the baseline sees results. Measured here, that makes them near-equal scorers that disagree on roughly a third of matches — neither dominates because neither is given what the other knows. Section 4.6 adds as-of-date history features so the ensemble sees both.
- **A protected evaluation set.** Roughly fifteen configurations were compared over the course of this work against the same five random splits. That is model selection on a test set. Section 4.11 withholds a full season so the headline number has never been optimised against.
- **A genuine forward test.** Section 5.11 scores the model against the 2026 World Cup knockout bracket, which no part of the pipeline has seen.

Objective. Build a generalizable, player-aware ML model for World Cup match prediction and benchmark it against a historical win-ratio baseline.

Scope. FIFA World Cup matches (finals and qualifiers) using player rosters from FIFA game editions 15 through EA FC 25 (calendar years 2015–2025), framed as binary win/loss classification with draws excluded, consistent with the reference paper.

3 Literature Survey

Football outcome prediction has been approached with a wide range of methods, which broadly fall into three families:

- **Statistical / ranking-based models** (e.g. Elo, Poisson goal models) estimate team strength from historical score lines. They are simple and well-calibrated for frequently-played matchups, but weak whenever squad composition changes faster than the rating can adapt — exactly the World Cup case. Elo in particular remains a strong benchmark and, as Section 4.6 shows, a strong *feature*.
- **Team-level machine learning models** replace the hand-built rating formula with a learned classifier over team-level features (rankings, recent form, home advantage). These improve on pure statistical models but still treat the team as a single opaque unit.
- **Player-aware machine learning models** aggregate individual player attributes (technical, physical, mental, goalkeeping) into a team representation, allowing the model to react to roster changes between tournaments. This is the family this project belongs to, and the one followed

most directly by Al-Bustami & Ghazal (2025), whose feature-selection principles (relevance, completeness, non-redundancy), Weighted Win Ratio baseline, and five-model majority-voting ensemble this project reproduces.

The key gap the player-aware family addresses is **squad turnover**: a national team’s roster is rebuilt from players across different clubs every cycle, so a team-level historical record answers “how has this team name performed” rather than “how good are the players who will actually play.” Player-level aggregation closes that gap at the cost of needing per-edition player attribute data, which this project sources from the same family of public datasets used in prior player-aware work.

4 Methodology

4.1 Data Sources

Three public datasets were used — two player sources and one match source:

Level	Dataset	Source
Player	FIFA Complete Player Dataset, one file per game edition	Kaggle: stefanoleone992/fifa-{edition}-complete (FIFA 15–23, EA FC 24)
Player	EA Sports FC 25 database	Kaggle: nyagami/ea-sports-fc-25-database-rating
Match	International football results, 1872–present	Kaggle: martj42/international-football-results- (slug frozen, dataset actively updated)

The player dataset is versioned per game edition, mapped to a roster year via `edition = year - 2000`, covering 2015–2025. EA renamed the series after FIFA 23 and the original author stopped publishing after EA FC 24, so the 2025 edition comes from a different author with its own schema and is normalized onto the canonical column set by `src/data/adapters.py`. Four attributes (`value_eur`, `wage_eur`, `international_reputation`, `potential`) are simply absent from that source and are therefore excluded from *every* year, rather than being imputed for 2025.

The match dataset provides **49,518 played international matches**, of which **9,839** are World Cup finals or qualifiers. Of those, **2,730** fall in the 2015–2025 player-data window; **581 (21.3%)** are draws and excluded by the binary framing, and **1,504** of the remaining 2,149 survive the join to squad profiles for both sides. The full international record — not just the World Cup subset — is used to compute the match-history features of Section 4.6, since Elo and form need volume to mean anything.

4.2 Squad Selection

Each player record carries a `nation_position` field, populated only for players EA has licensed as part of that edition’s official national squad. Because EA licenses only a subset of federations per edition (50 nations in FIFA 18 versus 33 in FIFA 22), using `nation_position` alone would silently drop the unlicensed federations’ matches — 13 of 40 teams from the 2018 and 2022 World Cups, leaving 35 of 128 matches usable instead of 118.

Squad selection therefore has two paths: the licensed call-up list where a (nationality, year) pair has one, and a **proxy squad** picked by rating where it does not. The proxy path fills a positional quota — 3 goalkeepers, 8 defenders, 8 midfielders, 4 forwards, backfilled from the best remaining players — rather than taking the top 23 by **overall** outright.

That distinction is not cosmetic. EA rates goalkeepers systematically below outfielders, so a nation’s best keeper often falls outside its top 23 entirely (Algeria 2017’s ranks 32nd among his compatriots). Taking the top 23 outright left **58% of proxy squads with no goalkeeper at all**, which skews every squad mean toward attacking attributes and makes any position-restricted aggregate meaningless. The quota fixes both.

A consequence worth stating plainly: the FC 25 source carries no national-team tags at all, so every 2025 squad is a proxy. This is a difference of degree rather than kind — the fallback was already load-bearing for unlicensed federations in earlier years — but it means the holdout season is built entirely from proxy squads.

4.3 Data Cleaning

Following the reference paper’s three feature-selection principles:

1. **Relevance** — only attributes describing offensive, defensive, physical, mental, or goalkeeping proficiency are retained.
2. **Completeness** — columns missing for more than 20% of rows are dropped, applied *after* squad filtering, since `nation_position` is itself null for ~94% of the unfiltered player pool by design.
3. **Non-redundancy** — highly correlated attributes (pairwise correlation > 0.9) can be dropped to avoid multicollinearity.

4.4 Team-Year Profiles

Individual player rows are aggregated into a single **team-year profile** per (nation, year) pair, producing **1,749 profiles across 189 nations** — one row per national team per edition, so the same team carries a different feature vector in different cycles. Median squad size is 16 rated players.

Aggregation is deliberately of three kinds, because a flat squad mean answers the wrong question in two places:

- **Squad means** over every numeric attribute — the paper’s original aggregation, and a fair summary of squad *depth*.
- **Shape of the overall rating** (`overall_max`, `overall_top3`, `overall_spread`). The mean measures depth, which wins leagues; a single knockout match is often decided by the two or three best players on the pitch. Norway 2025 is the worked example: squad mean 76.6 against Brazil’s 84.4, but a higher best player (91 to 90) and a top-3 within 1.7 points. By the mean it is a mismatch; by the shape it is the close game it turned out to be.
- **Position-restricted summaries** (`overall_gk/def/mid/att`, `goalkeeping_core`, `defending_core`, `attacking_core`). Averaging `goalkeeping_*` across 23 players, ~20 of whom are outfielders with junk keeper ratings, measures almost nothing. These are added *alongside* the squad-wide means rather than replacing them, because a nation whose rated pool contains no keeper yields NaN here — true of 27% of match rows.

4.5 Feature Engineering

Each match is between a home team (“Team A”) and an away team (“Team B”). For every profile attribute the feature row carries the difference `attr_diff = attr_a - attr_b`.

Only the difference is kept. The `_a`, `_b`, `_diff` triple the paper implies is collinear by construction — `diff` is exactly `a - b` — and costs three times the dimensionality for no extra information about which side is stronger. Measured across five seeds, dropping to differences alone (188 features $\rightarrow$ 64) held accuracy while cutting seed-to-seed spread from 1.9 to 1.1 points.

Two features describe the *fixture* rather than either team and so carry no suffix:

- **neutral_site.** World Cup qualifiers are played home-and-away and the home side wins **63.6%** of them; on neutral ground it is **46.8%**, near a coin flip. Without this flag the model blends the two and applies a phantom home advantage to every neutral-venue fixture — which is every match at the tournament itself.
- **squad_size_min.** Profiles are assembled from whoever EA rated, so a “team” can be two players: 47.5% of match rows have at least one side built from fewer than 11 rated players, and 32.6% from fewer than five. The minimum is the right summary, since a comparison is only as reliable as its weaker half.

Squads are joined by match date, not calendar year. The edition labelled year *Y* ships in about September of *Y-1*, and **67.6%** of the matches in scope are played September to December. Joining on the calendar year hands those fixtures ratings a median of twelve months old while a fresher edition already exists. Joining on the edition current at the match date moves 55% of rows onto newer squads, and was worth **+1.5 points** — the single largest gain of any change in this project (Section 5.13).

This produces **64 features** per match. Draws are excluded, giving 1,504 usable matches.

4.6 Match-History Features

The player-attribute features describe squads and say nothing about results; the WWR baseline is the mirror image. Measured directly, the two scored within a point of each other while disagreeing on roughly a third of matches, and on the matches the baseline got wrong the ensemble was right 44% of the time — a perfect router between them would have scored 85.9%. The ensemble had simply never been given what the baseline knows.

`src/features/history_features.py` closes that gap with six per-team families, each emitted as a difference:

Feature	Meaning
<code>elo_diff</code>	Classic Elo, updated chronologically over the whole record
<code>wwr_diff</code>	The baseline’s own credibility-weighted win ratio
<code>played_diff</code>	Matches played to date (the <i>v</i> term WWR shrinks on)
<code>h2h_winrate_diff</code>	Head-to-head record against this specific opponent, shrunk toward 0.5
<code>form_win_diff</code>	Win rate over the last 10 matches
<code>form_gd_diff</code>	Mean goal difference over that same window

Every value is computed as of the match date, from matches strictly earlier than it. Leak-freedom is therefore structural rather than a property of the split: the frame stays honest under any split strategy, including the random split used here, where a train-split-only fit would still let a 2022 result inform a 2018 prediction. Same-day fixtures are excluded too — every match on date D is snapshotted before any of D ’s results are applied, so two matches on the same day cannot see each other regardless of file order. The property is invisible in the output (a leaky version produces a perfectly plausible frame), so it is asserted directly in `tests/test_history_features.py`.

Elo weighting was implemented following the World Football Elo convention and then split in two when measured separately — and only half of it survived. See Section 5.13.

4.7 Training-Row Mirroring

World Cup finals are played on neutral ground, but the source data still lists one team as “home”, and 61% of rows are home wins. Without intervention the model learns argument order as if it were signal. Mirroring appends a side-swapped copy of a training row with the label flipped.

Two constraints make it work rather than hurt:

- **Neutral rows only.** Mirroring *everything* cost about two points of accuracy, because asserting that the away side of a qualifier had home advantage is simply false. Restricted to rows where the side ordering really is arbitrary, it adds 0.4 points and cuts seed-to-seed spread from 3.0 to 2.0.
- **Strictly after the train/test split.** Mirroring first would put a row and its own copy on opposite sides of the split — the same match in both train and test.

4.8 Baseline Model: Weighted Win Ratio (WWR)

The baseline uses only team-level historical results, per the reference paper’s Section IV:

- If two teams have played at least 5 matches head-to-head (the paper’s 75th-percentile threshold), the team with more head-to-head wins is predicted to win.
- Otherwise, each team’s **Weighted Win Ratio** is a credibility-weighted shrinkage estimate:

$$\text{WWR} = \frac{v}{v+m} R + \frac{m}{v+m} C$$

where v is the number of matches the team has played, R its win ratio, C the global win ratio across all teams, and m a shrinkage parameter (set to 10). This is the same family of estimator used for IMDB’s weighted ratings: teams with few recorded matches are pulled toward the global average, teams with a long track record are trusted closer to their own ratio.

The baseline is fit on the **full 9,839-match World Cup history** — all years, not merely the 2015–2025 window with player data — excluding only the exact rows held out for testing in a given run, matched on `(date, home_team, away_team)`. It is therefore a strong benchmark rather than a straw man, and Section 5.12 tests directly whether it is handicapped relative to the ensemble.

4.9 Ensemble Model

Five classifiers are trained and combined via **hard majority voting**:

Model	Role
Logistic Regression	Linear, interpretable baseline-within-the-ensemble
Random Forest	Tree-based, robust on tabular features
XGBoost	Gradient-boosted trees, strong on tabular data
AdaBoost	Boosted weak learners, a different bias/variance trade-off
K-Nearest Neighbors	Non-parametric, adds a different decision boundary

Each model’s hyperparameters are selected by 5-fold cross-validation on the training split (grid search per model type) before being wrapped in a `VotingClassifier`. Both the CV fold shuffle and each estimator’s own `random_state` are seeded from the current run’s seed — holding model randomness fixed across “different” seeds would make the reported spread reflect split variance alone.

4.10 Evaluation Protocol

Each run performs a stratified 80/20 train/test split and reports overall accuracy, accuracy split by high- versus low-scoring matches (combined goals ≥ 3), and accuracy on “challenging cases” — the subset of test matches the baseline got wrong, isolating whether player-level features add value precisely where team-level history fails.

Four operations are fit on the training split only, never the full frame: missing-value imputation, the scaler, PCA, and the hyperparameter search. Row mirroring likewise happens after the split. Every reported development number is averaged across **five random seeds** (0, 1, 17, 42, 123) with the standard deviation alongside.

4.11 Why There Is a Holdout

Roughly fifteen configurations were compared against those same five random splits over the course of this work, each kept or discarded on how it scored there. That is model selection on a test set, and its effect compounds quietly. It showed up directly: across the same stretch the random-split figure climbed 78.9% $\rightarrow$ 80.3% while an untuned temporal split moved 79.7% $\rightarrow$ 79.5%. The last round of apparent gains had not replicated.

The 2025 season is therefore withheld entirely. `build_match_dataset()` drops it before returning, so it reaches neither feature selection, hyperparameter tuning, the seed-variance charts, the figures, nor the served model. The only reader is `python -m src.holdout`, which trains on all 1,223 development matches and scores the 281 withheld ones exactly once.

The cost is real — 281 fewer training matches — and the benefit is a number no decision was made against. One caveat is kept explicit in Section 6: the holdout was locked *after* most configuration choices had already been made with 2025 visible, so it is clean from the FC 25 adapter onward, not for the whole history of the project.

5 Results and Evaluation

Results are reported at two levels, and the distinction matters: **development numbers choose the model, the holdout number reports it.** They are not comparable row-to-row, and are never quoted interchangeably below.

5.1 Headline: the Withheld 2025 Season

Trained on all 1,223 development matches (2015–2024) and scored once on the 281 withheld matches of 2025:

Model	Accuracy
Proposed ensemble	82.2%
Weighted Win Ratio baseline	75.4%
Always predict the home side	59.4%

The **+6.8 percentage-point margin is statistically significant** — McNemar’s test on the paired predictions gives $p = 0.0134$ ($b = 36$ matches the ensemble won and the baseline lost, against $c = 17$ the other way) — which is the first time that has been true in this project.

The supporting metrics on the same 281 matches:

	Ensemble	Baseline
Overall accuracy	82.2%	75.4%
High-scoring / low-scoring	86.1% / 77.2%	78.5% / 71.5%
Precision / recall / F1	0.842 / 0.862 / 0.852	0.827 / 0.743 / 0.782
ROC-AUC	0.907	—
Challenging cases recovered	36 of 69 (52.2%)	—
Unanimous vote (5/5)	85.1% on 242 matches	—
Split vote (4/5 or 3/5)	64.1% on 39 matches	—

Individual base models score 79.4% to 82.6% before the vote, so the ensemble sits at the top of its own members’ range rather than beyond it.

Two qualifications belong next to that number rather than in a footnote. First, all 281 holdout matches are World Cup **qualifiers**; no finals were played in 2025, so the headline is measured on qualifying fixtures only. Second, the holdout was locked partway through the project, as Section 4.11 describes. That it came in *above* the development estimate rather than below is the reassuring direction, but it is a single measurement of a single season.

5.2 Development Results Across Five Seeds

Every remaining number in this section comes from development data (2015–2024, 1,223 matches, 245 per test split), averaged over five random seeds:

Model	Overall Accuracy
Proposed ensemble	79.0% $\pm$ 3.3%
Weighted Win Ratio baseline	76.0% $\pm$ 3.1%

Per seed:

Seed	Ensemble	Baseline	Margin
0	83.7%	80.0%	+3.7 pp
1	76.3%	72.7%	+3.7 pp
17	81.2%	78.0%	+3.3 pp
42	76.3%	73.5%	+2.9 pp
123	77.6%	75.9%	+1.6 pp
Mean	79.0%	76.0%	+3.0 pp

The ensemble is ahead on **all five seeds**. That consistency is the meaningful claim at this sample size; the $\pm 3.3\%$ spread means the *level* on any single split should not be over-read.

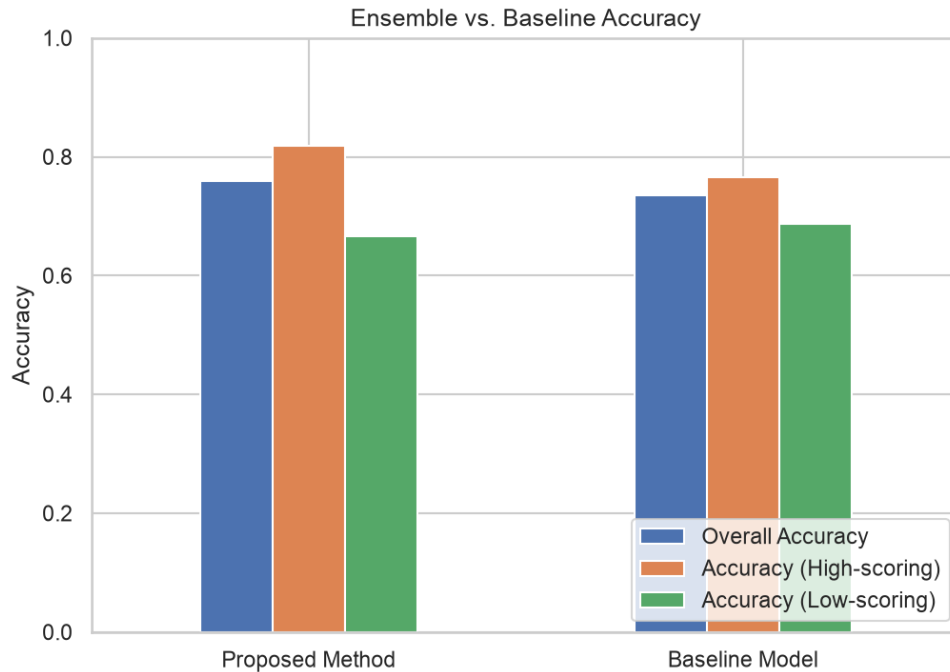


Figure 1: Ensemble vs. baseline accuracy, overall and split by high/low-scoring matches (seed 42).

5.3 High- vs. Low-Scoring Matches

Splitting by combined goals (threshold 3) tests whether either model's edge depends on how decisive a match was. Averaged over five seeds:

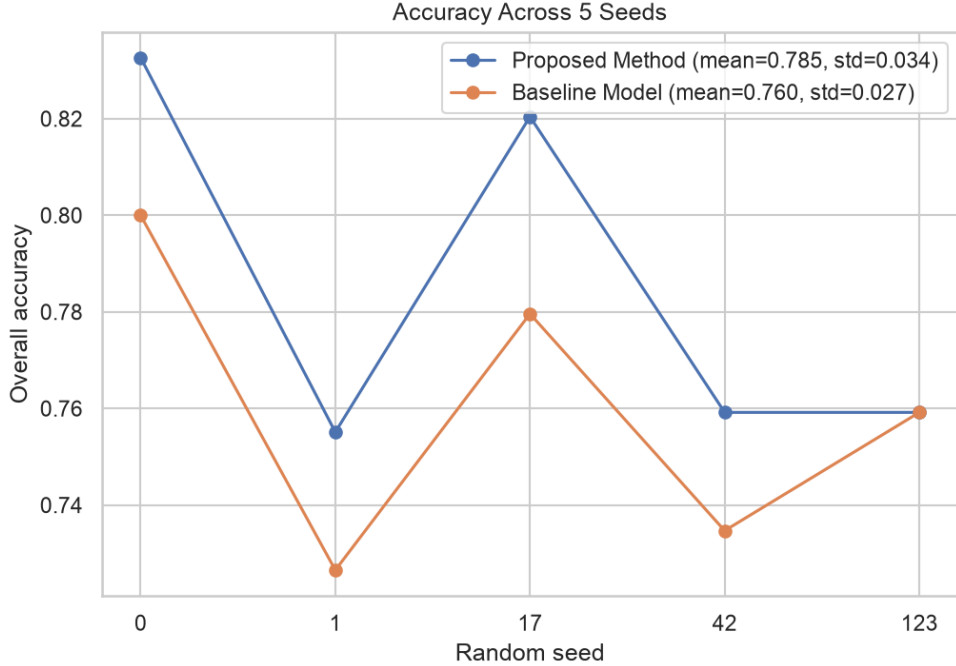


Figure 2: Overall accuracy of the ensemble and the baseline across five random seeds.

	High-scoring	Low-scoring
Proposed ensemble	83.4%	73.3%
WWR baseline	80.2%	70.6%

Both models find tight matches harder, by roughly ten points, and the ensemble’s advantage holds in both buckets — so the edge is not an artifact of blowouts being easier to call.

5.4 Challenging Cases

Isolating the test matches the baseline got wrong — 294 of 1,225 pooled predictions, or 24% — the ensemble recovers **41.8%** of them. The reference paper reports 25.0% (7 of 28) on the same metric. This is the most direct evidence that the player-level and history features carry signal precisely where team reputation fails.

5.5 Precision, Recall, F1, ROC-AUC

Home wins outnumber away wins 61% to 39%, so accuracy alone can mask asymmetric error rates. Averaged over five seeds:

	Precision	Recall	F1	ROC-AUC
Proposed ensemble	0.802	0.873	0.836	0.871
WWR baseline	0.835	0.757	0.794	—

The ensemble trades a little precision for substantially higher recall — it is more willing to call a home win, and right often enough when it does that F1 improves by over four points. ROC-AUC is computed from the mean soft-vote probability, since a hard vote emits no scores.

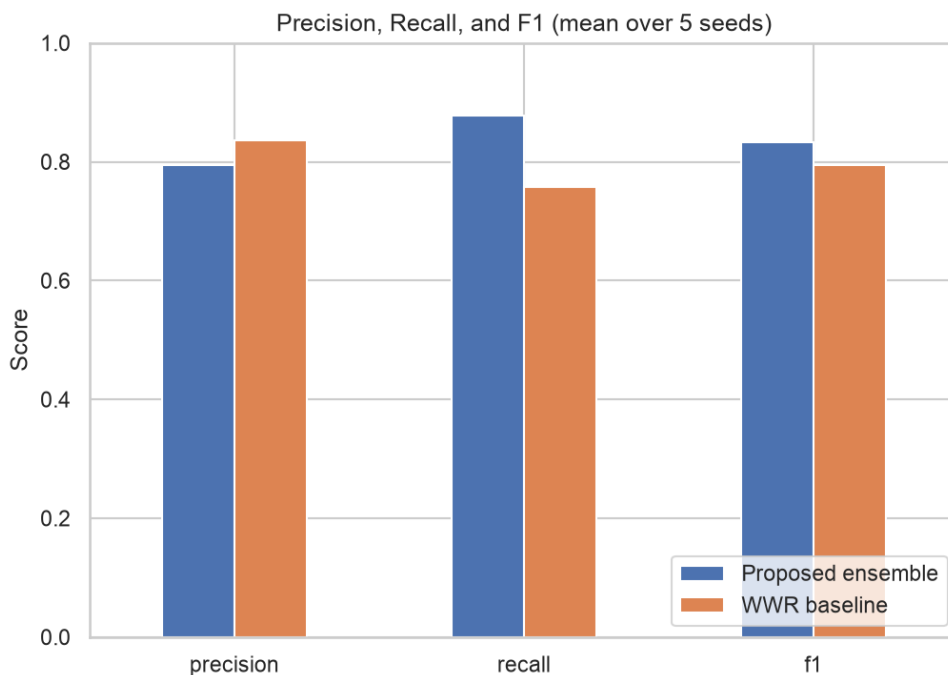


Figure 3: Precision, recall, and F1 for the ensemble and the baseline, averaged over five seeds.

5.6 Statistical Significance

McNemar’s test was run on the paired baseline-versus-ensemble predictions for every seed. On development splits the per-seed p -values are **0.188, 0.272, 0.230, 0.391 and 0.617** — **none reaches** $p < 0.05$. This is reported plainly rather than omitted: with 245 test rows and a three-point gap, the test is simply underpowered, which is exactly why this project averages over five seeds instead of reporting one run.

The holdout is where the evidence becomes statistical rather than directional. Training on the full development set and scoring 281 unseen matches, the larger +6.8-point margin reaches $p = 0.0134$. The honest summary is therefore: the development margin is consistent in direction on 5 of 5 seeds but not individually significant on any of them; the holdout margin is significant.

5.7 Per-Model Accuracy and Vote Agreement

Individual base-model accuracy ranges from 75.1% to 83.7% across models and seeds, overlapping the ensemble’s own range — no single model dominates, so majority voting is combining comparable classifiers rather than averaging one good model into four bad ones.

Splitting ensemble accuracy by how many of the five models agreed, pooled over all five seeds:

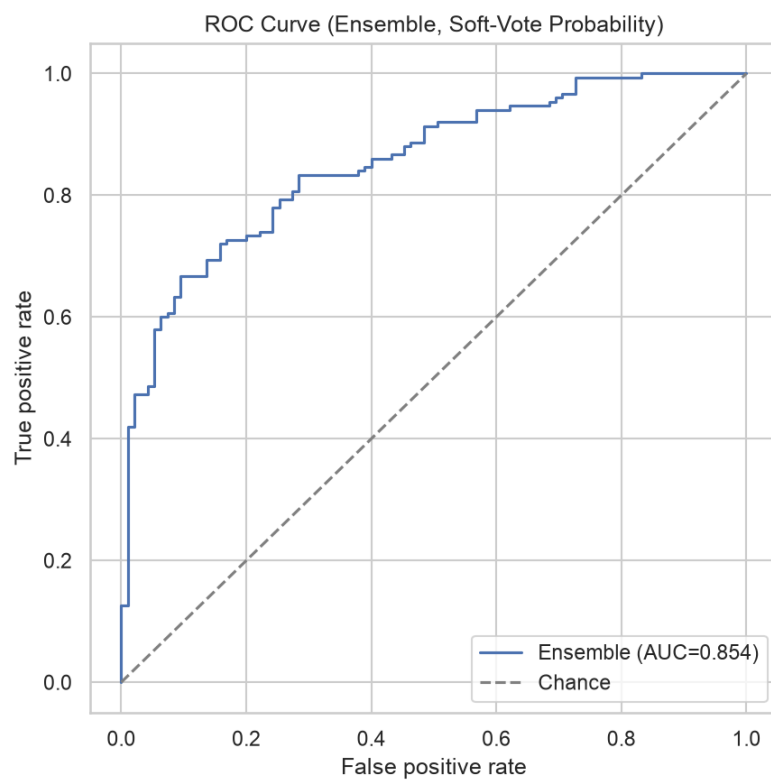


Figure 4: ROC curve for the ensemble's soft-vote probability, seed 42 (AUC 0.858).

Vote	Matches	Accuracy
Unanimous (5/5)	1,008	83.9%
Split (4/5 or 3/5)	217	56.2%

Vote agreement is therefore a usable **confidence signal** at inference time — a unanimous call is right five times in six, a contested one barely better than a coin flip. This is the “model agreement” indicator surfaced in the web app’s prediction view.

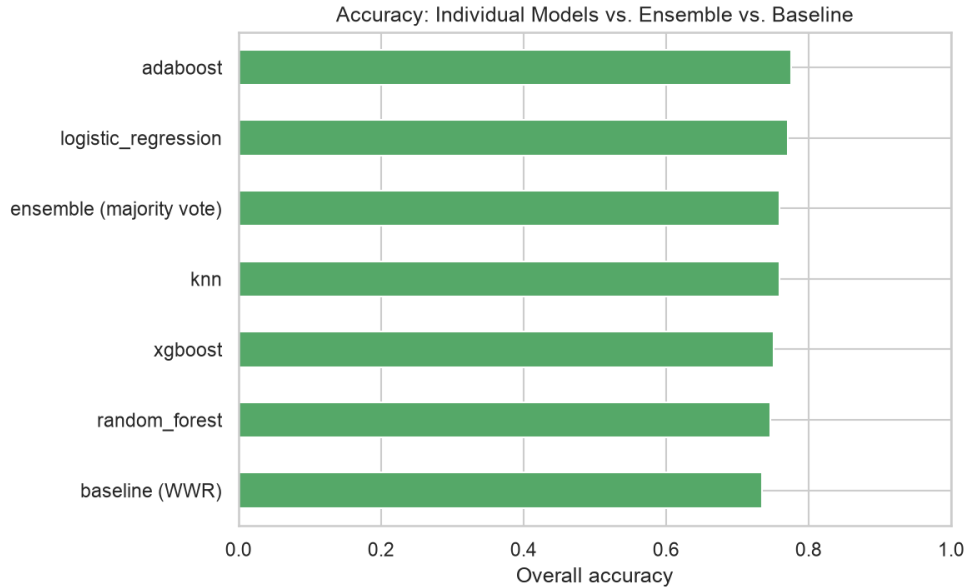


Figure 5: Accuracy of each individual base classifier vs. the full ensemble and the baseline, seed 42.

5.8 Feature Importance

Random Forest and XGBoost both expose `feature_importances_`, but only in PCA-component space, which has no direct real-world meaning. Mapping each component’s importance back to the original 64 engineered features — weighted by that feature’s *squared* PCA loading (the variance-share form; loadings sum to 1 per component this way), then renormalized — gives a consistent ordering across seeds. An earlier version of this mapping used the absolute loading instead, which over-credits features with many small, weakly-loading contributions; that flattened the whole top-15 into a visually indistinguishable band (0.018–0.024 across all 15) and buried the two strongest features entirely. Squared loadings recover a sharper ranking.

`squad_size_min`, `neutral_site` and `overall_spread_diff` rank in the top three on **every** seed (the first two trade rank 1/2 by seed; the third is always third) — match-context and squad-depth signals, not squad-quality comparisons. `squad_size_min` is how much roster actually backs the comparison (thin squads mean less signal either way); `overall_spread_diff` is the gap in how evenly rated each squad is top-to-bottom (star-studded-and- thin vs. uniformly solid); `neutral_site` captures the home-advantage asymmetry documented in Section 4.5. Immediately behind them, `played_diff`, `h2h_winrate_diff` and `age_diff` are the next-most-consistent block across seeds, so the match-history features added in Section 4.6 still carry real weight — they’re

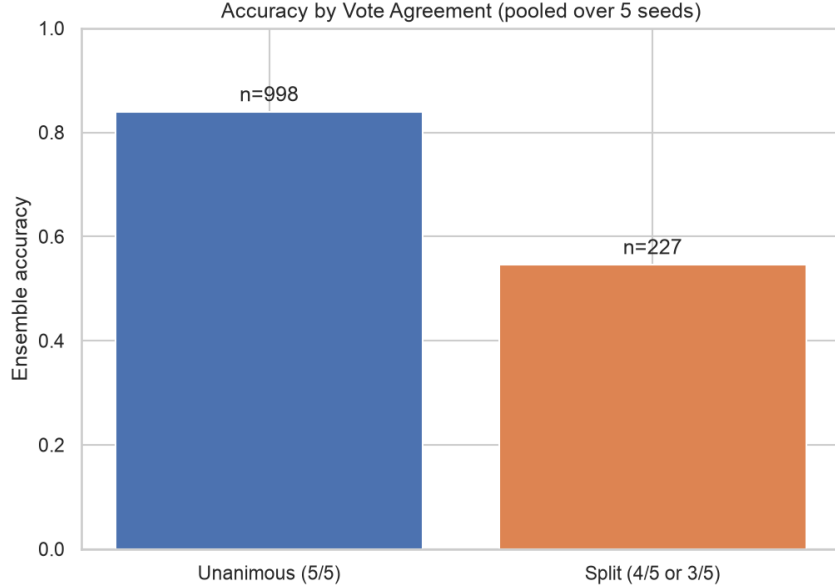


Figure 6: Ensemble accuracy by vote agreement (unanimous vs. split), pooled over five seeds.

just not the single dominant signal the way they first appeared to be. Among the squad attributes, `power_jumping_diff`, `weak_foot_diff`, `skill_moves_diff` and the position-restricted `attacking_core_diff` / `overall_mid_diff` rank highest — physical profile and technical versatility rather than headline ratings.

Two caveats. The importances still spread broadly rather than isolating one standout feature — expected when summing loadings across ~22 components — so this is a rough ordering, not a precise attribution. And `age_diff` ranking this high is more plausibly a proxy for squad experience and depth than a causal claim about age.

5.9 Dimensionality Reduction

PCA retains **22–23 components of the 64 standardized features** at the 95% variance threshold, depending on seed. The ratio is much higher than under the original `_a/_b/_diff` representation, because most of the redundancy PCA used to absorb there was collinearity introduced by the feature construction itself, and has now been removed at source (Section 4.5). What survives is correlation between genuinely distinct attributes — sprint speed and acceleration, or finishing and shot power — which is real structure in the data rather than an artifact of how the row was built.

5.10 Comparison with the Reference Paper

Metric	This project (holdout)	This project (development)	Al-Bustami & Ghazal (2025)
Overall accuracy	82.2%	79.0% ± 3.3%	59.38%
Baseline accuracy	75.4%	76.0% ± 3.1%	56.25%
Margin	+6.8 pp	+3.0 pp	+3.13 pp
Significant?	yes ($p = 0.0134$)	no (5/5 in direction)	not reported

Metric	This project (holdout)	This project (development)	Al-Bustami & Ghazal (2025)
Challenging cases	—	41.8%	25.00%
Test set size	281	245 per seed	96

The development margin reproduces the paper’s headline contribution almost exactly (+3.0 against +3.13 points) at roughly twenty points higher absolute accuracy. The holdout margin is more than double it, on nearly three times the test data, and is statistically significant where a three-match difference out of 96 could not be.

It is worth being precise about why the *margin* is not larger despite the absolute accuracy being so much higher: the baseline here is far stronger. A WWR baseline at 75.4% on binary no-draw outcomes is a hard reference, where the paper’s sits at 56.25%, barely above chance. Expressed as error reduction, the holdout ensemble eliminates **27.6%** of the baseline’s errors (24.6% $\rightarrow$ 17.8%); the development ensemble eliminates 12.5%.

5.11 Held-Out Test: the 2026 World Cup Knockout Bracket

The match dataset contains the 2026 World Cup, but `data.years` stops at 2025, so no 2026 fixture reaches the ensemble or the baseline. Team strength comes from EA FC 25 squad ratings published before the tournament began on 2026-06-11, and the history features are cut at that same kickoff date. A 2026 result therefore cannot inform a 2026 prediction:

Round	Ties	Ensemble	Baseline
Round of 32	16	88%	50%
Round of 16	8	88%	88%
Quarter-finals	4	100%	100%
Semi-finals	2	0%	50%
Overall	30	83.3%	66.7%

Every tie is scored with neutral-venue symmetrisation — the match is evaluated in both orientations and averaged — so argument order cannot decide the winner.

Three caveats keep this honest. With $n = 30$ the 95% interval is $\pm 13.3\%$, so the 17-point margin over the baseline is **not** statistically significant. Four of the thirty ties were decided on penalties, which the model has no way to represent: it predicts a 90-minute winner and is scored against the shootout result. And 83.3% sitting above the development figure is a warning sign as much as a triumph — the goal-difference Elo change was worth +0.33 points in cross-validation and appears to be worth +10 here, a gap of three matches out of thirty that is mostly luck. The cross-validated figure is the one to quote.

The model misses both semi-finals, picking France over Spain and England over Argentina. Neither the third-place playoff nor the final has a recorded score, making both genuine forward predictions: **England** third (62%), and **Spain** over Argentina in the final (52%) — barely above a coin flip, which is the correct amount of conviction for that fixture.

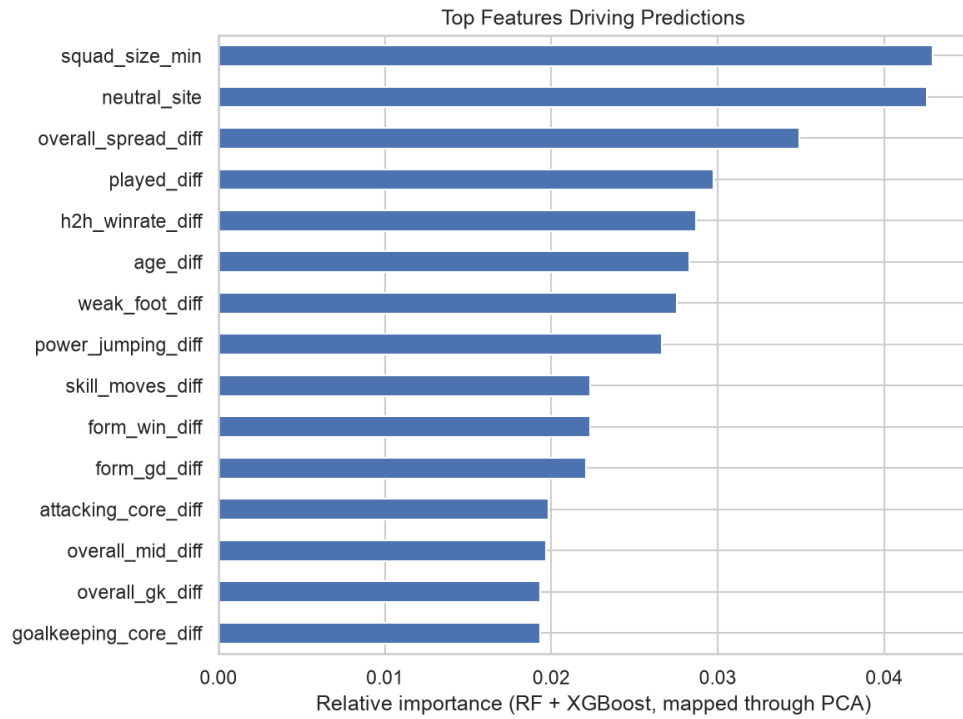


Figure 7: Top 15 engineered features by importance, mapped back from PCA-component space (seed 42).

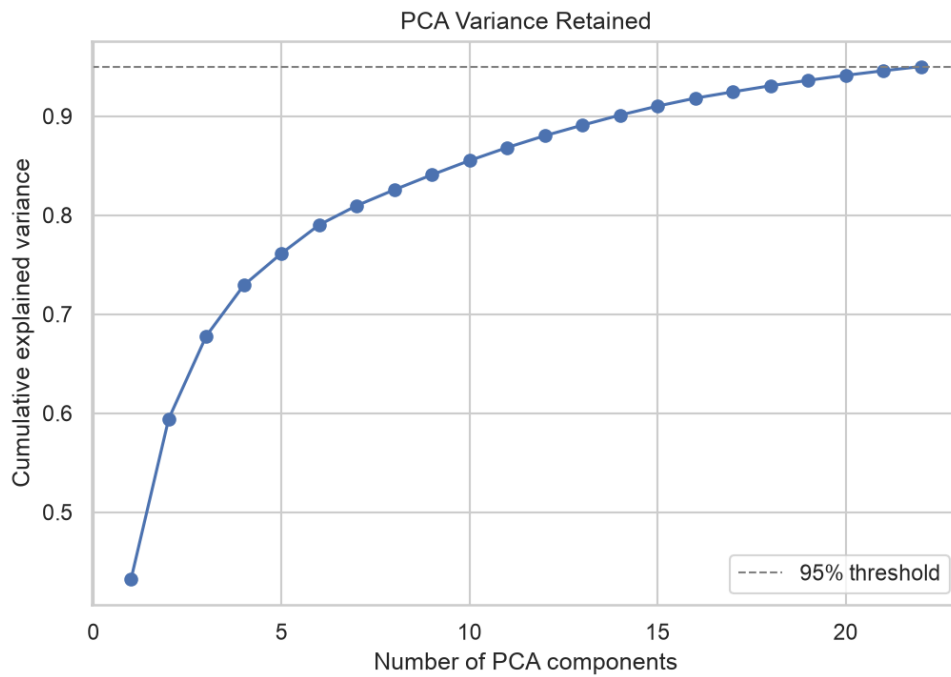


Figure 8: Cumulative explained variance vs. number of PCA components (seed 42).

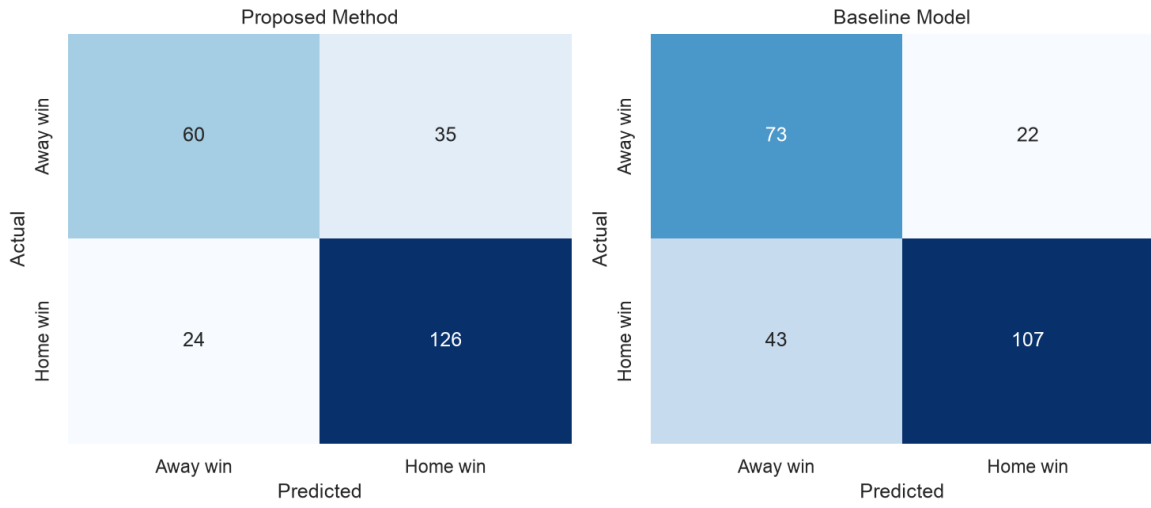


Figure 9: Confusion matrices for the proposed ensemble (left) and the baseline (right), seed 42.

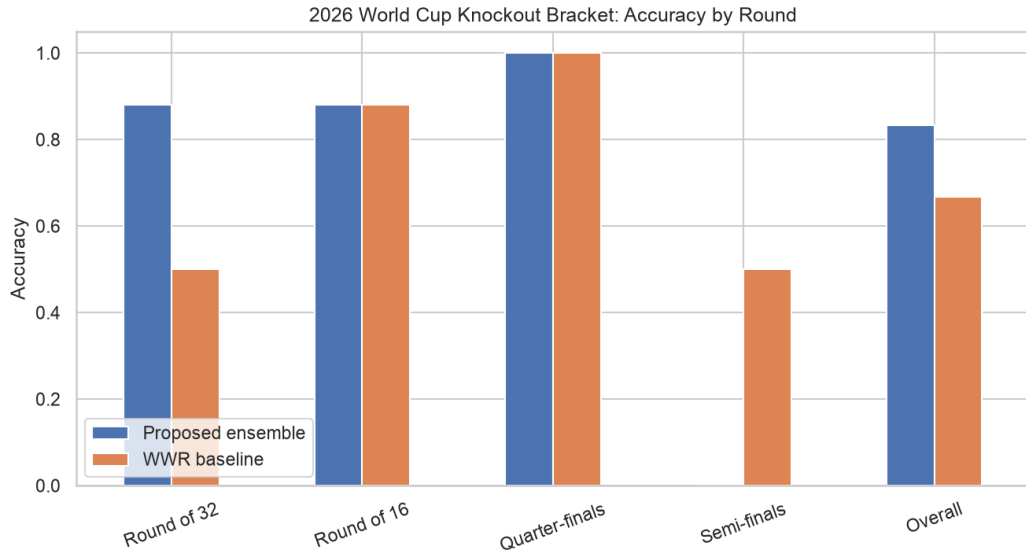


Figure 10: Ensemble vs. baseline accuracy per knockout round on the 2026 bracket, plus the overall figure.

5.12 Validity Checks

Two things that would invalidate the headline were tested directly rather than assumed. Both were measured at the configuration current when the check was run, before the holdout was locked, and are reported as such.

- **Is the random split hiding that the model trains on the future?** No. Splitting temporally instead — training on everything up to 2024-10-15 and predicting the 295 matches after it — gave **79.7% $\pm$ 0.7% against the baseline’s 77.0%**. Accuracy does not drop, so the random-split figure is not inflated by hindsight and the forecasting framing is defensible.
- **Is the baseline handicapped?** Slightly, and it was worth checking: the ensemble’s history features are built from all 49,518 international matches while the WWR baseline is fit on the 9,839 World Cup ones — a five-fold information edge. Giving the baseline the full record moved it from 74.8% to 74.9%, making the like-for-like margin +4.0 points rather than +4.2 at the point measured. Real, but too small to change the conclusion.

5.13 What Was Measured and Rejected

Every modelling change was ablated across all five seeds before being kept. Single-seed deltas of two to three points are routine noise at this sample size, and three of the changes below looked obviously right and were not.

Changes that were kept, each row cumulative on the one above it:

Change	Accuracy	Effect
Starting point (squad attributes only)	75.5% $\pm$ 2.5%	—
+ neutral_site feature	76.3% $\pm$ 2.3%	+0.8 pp
+ as-of-date history features	78.1% $\pm$ 3.0%	+1.7 pp
+ neutral-only row mirroring	78.5% $\pm$ 2.0%	+0.4 pp
+ squad shape & position aggregation	78.6% $\pm$ 1.8%	+0.1 pp
+ goal-difference-weighted Elo	78.9% $\pm$ 1.9%	+0.3 pp
+ _diff -only representation	78.8% $\pm$ 1.1%	-0.1 pp, spread 1.9 $\rightarrow$ 1.1
+ edition-aware squad join	80.3% $\pm$ 3.2%	+1.5 pp

Changes that were rejected, each measured against the configuration current at the time rather than against the cumulative total above:

Change	Effect	Reference
Competition-weighted Elo	-0.33 pp	78.26% vs flat-K 78.60%
Both Elo weightings together	-0.47 pp	78.13% vs flat-K 78.60%
Mirroring <i>every</i> training row	$\approx$ -2 pp	vs neutral-only mirroring
<code>min_squad_size</code> = 11	-0.9 pp	vs no filter
<code>min_squad_size</code> = 18	-3.1 pp	vs no filter

Every figure in both tables was measured on the full 1,504-match set *before* the holdout was locked, so none is comparable row-to-row with the 79.0% development figure, which is computed on the

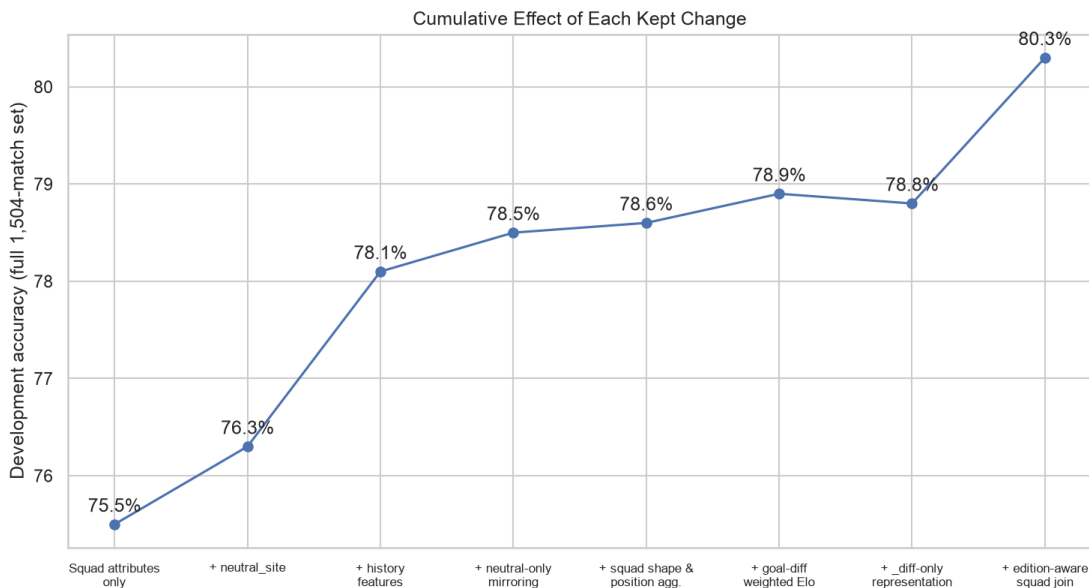


Figure 11: Cumulative development accuracy through each kept change, in the order it was added.

1,223 matches that remain after 2025 is withheld. The tables record which direction each change moved the model, not its level today.

Three of these deserve comment:

- **Competition-weighted Elo is the textbook choice and made things worse.** Setting K from the tournament (World Cup 60 ... friendly 20) does exactly what it promises — it halves an inflated Brazil–Norway Elo gap from +176 to +82 — but costs 0.3 points, because it makes every rating more reactive to recent results, which `form_win` and `form_gd` already measure. What the ensemble wants from Elo is stable long-run reputation. The goal-difference multiplier was kept: margin of victory is per-match evidence no other feature carries.
- **Mirroring every row cost about two points**, for the reason given in Section 4.7 — it asserts a falsehood about qualifiers.
- **Squad-size reliability did not work in either form.** Exposing `squad_size_min` as a feature moved nothing (-0.06), and *filtering* thin training rows was actively harmful: the lost volume costs more than the added noise. The feature is kept because it is one column that genuinely describes the row; the filter ships disabled.

6 Discussion

The results support the central hypothesis: player-level attributes aggregated into year-specific team profiles provide predictive signal beyond what a purely historical baseline captures. On the withheld season the margin is 6.8 points and statistically significant, and the ensemble recovers 41.8% of the matches the baseline gets wrong.

The more transferable finding is about **information parity**. The original framing pits a squad-aware model against a results-aware one and finds them nearly equal — which is what this project measured too, at the starting point. The two were not equally good; they were differently informed,

disagreeing on a third of matches with a perfect router scoring 85.9%. Giving the ensemble the baseline’s own signal as as-of-date features is what turned a coin-flip comparison into a consistent margin, and it accounts for more of the total gain than any player-attribute refinement did.

The second transferable finding is that **the join matters as much as the features**. The single largest gain here was not a new model or a new attribute: it was noticing that FIFA edition Y ships in September of $Y-1$ while two-thirds of fixtures are played September to December, so most matches were being scored against year-old squads. That is worth more than every squad-aggregation refinement combined.

Limitations.

- **The task excludes draws.** 21.3% of in-scope matches end level and are dropped. The model cannot express “too close to call”, and the reported accuracies describe only decided matches.
- **The holdout is qualifiers-only.** No World Cup finals were played in 2025, so the headline is measured entirely on qualifying fixtures. Finals are only 100 of the 1,504 usable matches overall — this pipeline is validated mostly on qualifiers and applied to finals.
- **Squads are proxies more often than not.** Where EA licensed no national squad the profile is a quota-filled top-23 by rating; for 2025 that is every nation. Profiles also take no account of formation, tactical role, injuries, or which eleven actually started.
- **Thin profiles are common.** 47.5% of match rows have at least one side built from fewer than eleven rated players. Filtering them made things worse, so they remain — with `squad_size_min` exposed so the model can discount them itself.
- **The ensemble is under-confident.** Measured on an earlier configuration of this pipeline over 1,495 pooled held-out predictions: when it reported 70–80% confidence it was right 80.9% of the time, and at 80–90% it was right 95.1%. The compression comes from averaging five probability outputs toward 0.5. Brier score is 0.150 against 0.237 for predicting the base rate, so the ranking is sound and the *displayed numbers* are the problem — a calibration step would fix them without changing a single prediction.
- **The offline and live models vote differently.** The evaluation in this report uses hard majority voting, per the paper. The live `/predict` endpoint uses the mean soft-vote probability instead, so that the returned winner and its confidence can never point in opposite directions — possible under a hard vote when a narrow 3/2 majority disagrees with the average probability.

7 System Implementation

The pipeline above is exposed through a small full-stack application: a FastAPI backend that owns the trained model bundle, and a React/Vite frontend that consumes it. This section describes how the two are wired together and what running the system looks like.

7.1 Backend API

A FastAPI application exposes three endpoints. `GET /teams` lists every (team, edition) pair that has a squad profile, which the frontend uses to populate its team pickers and to restrict the edition selector to years both chosen teams actually share. `POST /predict` scores a single matchup: it returns the ensemble’s winner and per-model votes, and, where the scoreline model is enabled, an expected-goals and outcome-probability breakdown from it as well. Because World Cup matches are played at neutral venues but the model is trained on rows carrying a nominal home side, a request can ask to have the fixture scored in both team orderings and averaged (the `neutral` flag,

true by default) — which removes the model’s otherwise real sensitivity to which team happens to be passed first. Winner and confidence are both derived from the mean soft-vote probability across the five models rather than a hard majority vote, so the two can never point in opposite directions, which a narrow 3/2 hard-vote majority combined with a separately-computed confidence could otherwise do; the offline figures throughout this report still use the paper’s hard majority vote, since that only affects this single live endpoint. `GET /evaluation` returns everything the results dashboard renders, computed from the cached model bundle with no request parameters.

All three endpoints depend on a module-level state populated once, during the application’s startup handler, by loading or training the model bundle. A cache miss takes several minutes, during which the server does not accept traffic in any meaningful sense; the cheapest endpoint that only succeeds once startup has finished — `/teams` — is what the startup script and the frontend both poll on.

7.2 Model Caching Strategy

Training the full pipeline — five classifiers across five seeds plus the scoreline model — is expensive enough that it should not happen on every server restart. The caching layer hashes the training-relevant sections of the configuration file (data, features, baseline, models, evaluation, score-model settings) together with the fixed seed list and the active random seed into a short key, and reuses a previously trained bundle from disk whenever that key has not changed; changing an unrelated setting, such as the project’s display name, does not invalidate the cache, while changing any modelling knob does.

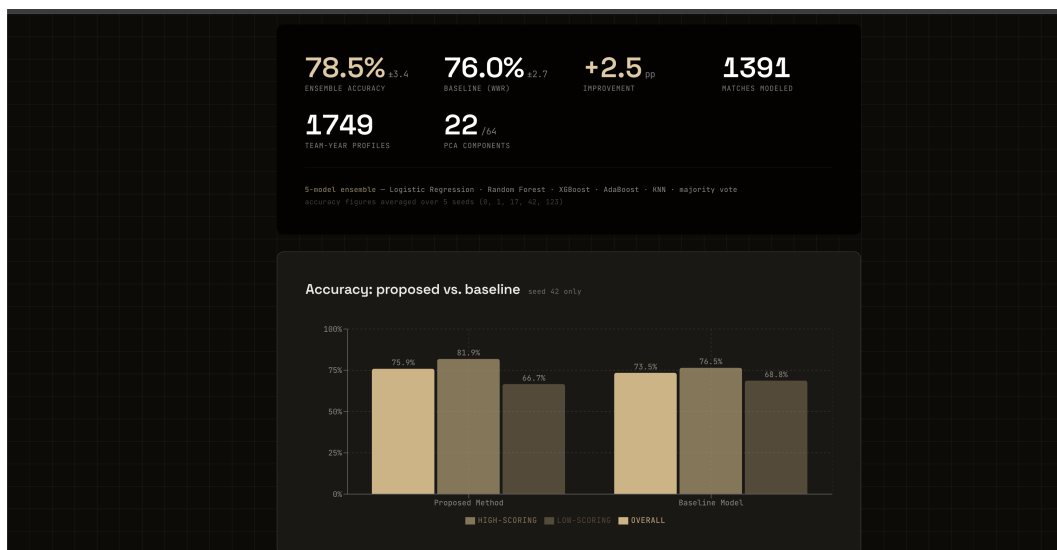
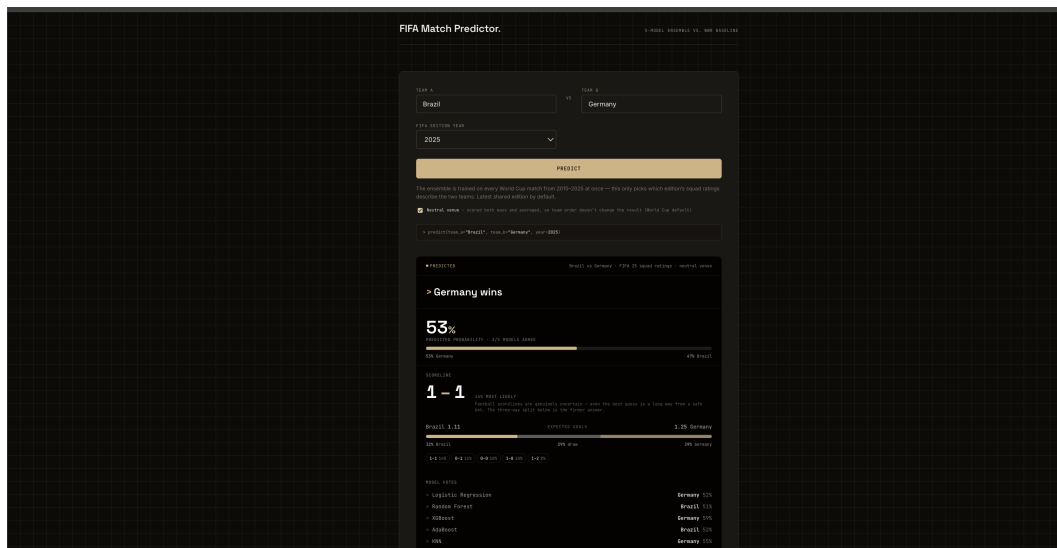
7.3 Frontend Application

The web application is a React and Vite single-page app with two panels: a live team-versus-team predictor and an evaluation-results dashboard, both driven entirely by the three API endpoints described above. Because the backend can take several minutes to finish training on a cache miss, the frontend treats a server error without an accompanying error message as “still starting” rather than as a failure, and polls until the backend responds properly or returns a genuine, message-bearing error such as an unknown team name.

The evaluation dashboard renders the same figures reported throughout this document — the headline stat tiles, the accuracy-versus-baseline chart, the seed-stability plot, PCA variance, per-model accuracy, the ROC curve, and feature importance — computed once at startup from the cached bundle rather than recomputed per request.

7.4 Running the System Locally

A single command (`python start.py`) starts both halves of the application, waits for the backend to finish training or loading from cache, and opens a browser window once the system is ready. The backend and frontend can also be run independently for development (`--backend-only` / `--frontend-only`), and anything already listening on the expected ports is reused rather than started twice, so a second invocation finds both ports taken and exits immediately. A companion script (`python stop.py`) kills by port to shut the servers back down, since that reuse behaviour otherwise leaves nothing for a plain Ctrl+C to interrupt.



8 Conclusion and Future Work

This project implements and validates a player-aware, ensemble-based World Cup match predictor, reproducing the reference paper’s methodology end-to-end on real public data and extending it in three ways: fusing the baseline’s own signal into the ensemble as leak-free as-of-date features, joining squads by the edition current on the match date, and withholding a full season so the headline number has never been optimised against.

On that withheld season the ensemble scores **82.2% against a 75.4% baseline — a significant 6.8-point margin** ($p = 0.0134$, **ROC-AUC 0.907**), eliminating 27.6% of the baseline’s errors. On development data it scores $79.0\% \pm 3.3\%$ against $76.0\% \pm 3.1\%$, ahead on all five seeds. Against the reference paper’s 59.38% / 56.25%, both the absolute level and the margin are substantially higher. The pipeline is wrapped in a FastAPI backend and React frontend giving live team-versus-team predictions with per-model agreement, alongside the evaluation dashboard that produced the figures here.

Future work, in descending order of expected value:

1. **Widen the training scope.** Training rows are restricted to World Cup finals and qualifiers — 1,504 usable matches — while the history features already read all 49,518 internationals. Training on every international in the window (10,680 matches, 8,224 decided) and continuing to *evaluate* on World Cup fixtures would roughly quadruple the training data. At the current sample size the $\pm 3.3\%$ seed spread makes any change worth less than about 1.5 points unmeasurable; more data is the only lever that both raises the ceiling and lowers that floor.
2. **Calibrate the probabilities.** Isotonic or Platt scaling on a held-out fold would correct the under-confidence documented above without changing the ranking, and would make the confidence shown in the web app mean what it says.
3. **Opponent-adjusted form.** Beating San Marino currently counts the same as beating France. A real defect, though the expected gain sits inside the present noise band — worth doing after (1), when it becomes measurable.
4. **Model draws.** A Poisson or bivariate-Poisson goals model would restore the 21.3% of matches currently discarded and let the model express genuine uncertainty, at the cost of changing the task so results are no longer comparable to anything above.
5. **Minutes-weighted aggregation** — weighting player attributes by starting-XI likelihood rather than a flat squad mean — and, as noted as optional scope in the project proposal, graph neural networks over player/team relationships if additional compute budget becomes available.

Any of these requires a new holdout season first: 2025 has now been scored, and tuning against it would turn it back into a development split.

9 References

Al-Bustami, A. & Ghazal, Z. “From Players to Champions: A Generalizable Machine Learning Approach for Match Outcome Prediction with Insights from the FIFA World Cup.” *2025 IEEE International Conference on Electro Information Technology (eIT)*, pp. 574–578. DOI: 10.1109/EIT64391.2025.11103598

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